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CS-499 Module 7 – Professional Self-Assessment

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Professional Self-Assessment

Introduction

Completing the Bachelor of Science in Computer Science program at Southern New Hampshire University has been a significant milestone in both my academic and professional development. Throughout the program, I strengthened my understanding of software development, algorithms, databases, security, and software engineering practices while also gaining practical experience applying these concepts to real-world projects. The development of this ePortfolio has provided an opportunity to reflect on my growth throughout the program and demonstrate how the knowledge and skills I have acquired can be applied to professional software solutions.

When I first began the Computer Science program, I viewed the degree primarily as a pathway into a software development role. However, as I progressed through my coursework and continued building my career as a Software QA Engineer, my perspective evolved. I discovered that the technical knowledge gained through computer science extends far beyond traditional software development and can be directly applied to software quality, automation, testing, and software architecture. As a result, my professional goals shifted toward becoming a more technical quality engineering professional and eventually pursuing opportunities as a Software Development Engineer in Test (SDET).

Collaboration and Communication

One of the most valuable lessons I learned throughout the Computer Science program is the importance of communication and collaboration within software development teams. Successful software projects require cooperation among developers, testers, project managers, stakeholders, and end users. Technical knowledge alone is not enough to deliver successful software solutions. Professionals must also be able to communicate ideas, document decisions,

explain technical concepts, and work effectively with individuals who have different responsibilities and levels of technical expertise.

My experience as a Software QA Engineer has reinforced the importance of collaboration. Quality assurance often serves as a bridge between technical teams and business stakeholders by communicating defects, validating requirements, and ensuring software meets user expectations. Throughout the program, I strengthened these communication skills through project documentation, written narratives, code reviews, presentations, and technical analyses. The code review included in this ePortfolio demonstrates my ability to analyze software, identify opportunities for improvement, and communicate enhancement plans in a professional and organized manner.

Software Engineering, Algorithms, Databases, and Security

The coursework completed throughout the Computer Science program provided experience across several core areas of computer science. Through software engineering projects, I learned how maintainable software systems are designed using principles such as modularity, separation of concerns, validation, and error handling. These concepts were demonstrated in my Software Design and Engineering enhancement, where I implemented Angular route guards, centralized authorization logic, improved validation, and strengthened application maintainability.

The Algorithms and Data Structures enhancement provided an opportunity to apply algorithmic thinking to solve practical problems. By implementing search, sorting, and pagination functionality within the application, I improved both usability and efficiency while demonstrating concepts such as collection processing, linear search, comparator-based sorting, and data organization. This enhancement reinforced the importance of selecting appropriate algorithms and understanding how data structures influence application performance.

The Database enhancement expanded the application's functionality through the implementation of a reservation system using MongoDB and Mongoose. This work included schema design, relationships between collections, CRUD operations, indexing, validation, and secure access to reservation data. Through this enhancement, I gained additional experience

designing database-driven applications and managing data integrity within a full-stack environment.

Security has also been a recurring theme throughout both my coursework and professional experience. Modern applications must be designed with security in mind from the beginning of the development process. Validation, authentication, authorization, and secure access to data all contribute to protecting systems from potential vulnerabilities. The route guard implementation, form validation improvements, JWT-protected endpoints, and user-specific reservation controls included in my enhancements demonstrate a security mindset that considers both current functionality and future risks.

Professional Growth and Career Goals

The Computer Science program has significantly expanded my technical abilities and strengthened my confidence in software engineering concepts. More importantly, it has helped me refine my professional goals. While I originally viewed the degree as preparation for software development positions, I now recognize how valuable these skills are within software quality engineering and automation-focused roles. The combination of quality assurance experience and computer science knowledge provides a strong foundation for advanced roles such as Software Development Engineer in Test (SDET), where software development, automation, testing, and quality all intersect.

The knowledge gained throughout this program will continue to support my professional growth by enabling me to contribute more effectively to software teams, design more sophisticated automation solutions, evaluate software architecture, and communicate technical concepts to diverse audiences. These skills will remain valuable as software systems continue to grow in complexity and organizations increasingly rely on automation, cloud technologies, and secure software development practices.

Artifact Summary

The artifacts included in this ePortfolio collectively demonstrate my growth throughout the Computer Science program and showcase the integration of multiple computer science disciplines within a single application. The code review established a foundation for evaluating the original CS-465 Travlr application and identifying meaningful enhancement opportunities.

The Software Design and Engineering enhancement improved maintainability, validation, security, and separation of concerns through the implementation of route guards and centralized authorization logic. The Algorithms and Data Structures enhancement introduced search, sorting, and pagination functionality that improved data processing and user experience. The Database enhancement transformed the application into a more complete travel reservation system by introducing reservation management, database relationships, indexing, and secure data access.

Together, these enhancements demonstrate my ability to analyze existing software, apply computer science principles, implement meaningful improvements, and communicate technical decisions effectively. The portfolio serves as evidence of both my academic accomplishments and my readiness to continue growing as a computer science professional. It reflects the knowledge, skills, and experiences gained throughout the program while supporting my future goals within software quality engineering, automation, and software development.